

TIGHT AND COMPACT UNIT COMMITMENT

A. Unit Commitment

The objective function, adapted from the CUC, follows from the CUC objective function through a direct substitution of the index, replacing the cluster index with the unit-specific index. Similar to the constraints in the previous section, the TC UC formulation incorporates technical constraints for each unit and system-wide constraints. However, these constraints are explicitly reformulated for the unit-based model and indexed by (g) to represent the unit.

1) Objective function

The objective function includes fuel costs, fixed operation costs, start-up costs, shutdown costs, upward and downward reserve costs, and load shedding costs. In this study, the fuel cost function is represented using a piecewise-linear cost function.

$$C^{TOT} = \sum_{t=1}^T \left[\sum_{g \in \Omega_G} \left(\sum_{s \in \Omega_{s,g}} (C_g^s \cdot p_{g,t}^s) + C_c^{fx} \cdot u_{c,t} + C_g^{SU} \cdot v_{g,t} + C_g^{SD} \cdot w_{g,t} + C_g^{R-up} \cdot r_{g,t}^+ + C_g^{R-down} \cdot r_{g,t}^- \right) + \sum_{i \in \Omega^L} Cls \cdot LDS_{i,t} \right],$$

2) Piecewise Linear Cost Modeling

Equations (2)-(3) are used to linearize the nonlinear fuel cost function of thermal units. Based on (2), the total power generation above the minimum output level is expressed as the sum of the power generated in each segment. In (3), the power generation limits of each segment are specified.

$$\sum_{s \in \Omega_{s,c}} p_{c,t}^s = p_{c,t}^*, \forall t, c \in \Omega_c \quad (2)$$

$$0 \leq p_{c,t}^s \leq (\bar{P}_c^s - \bar{P}_c^{s-1}) \cdot u_{c,t}, \forall t, c \in \Omega_c, s \in \Omega_{s,c}. \quad (3)$$

3) technical constraints of units

a) Logical constraints:

Using (4) and (5), the state transitions for each unit's commitment, start-up, and shutdown are enforced.

$$u_{g,t} - u_{g,t-1} = v_{g,t} - w_{g,t}, \forall t, g \in \Omega_G \quad (4)$$

$$v_{g,t} + w_{g,t} \leq 1, \forall t, g \in \Omega_G \quad (5)$$

b) Minimum up/down time constraints:

Minimum uptime and downtime constraints, as defined in equations (6) and (7), are derived from [1-3]. As demonstrated in [3], these equations offer the most efficient computational performance.

$$\sum_{\tau=t-MUT_g+1}^t v_{g,\tau} \leq u_{g,t}, \forall t, g \in \Omega_G \quad (6)$$

$$\sum_{\tau=t-MDT_g+1}^t w_{g,\tau} \leq 1 - u_{g,t}, \forall t, g \in \Omega_G \quad (7)$$

c) Generation constraints:

The maximum generation for each unit is imposed using (8), (9), and (10). Equations (8) and (9) define the maximum

generation for fast-start units (minimum up time=1), while (10) represents the maximum generation for the remaining units with longer minimum up time. However, equations (8) and (9) increase the tightness of the formulation [2]. Equation (11) ensures that each unit generates at least the sum of the downward reserve and minimum generation at each hour. Equation (12) states that each unit's generation equals its minimum generation level plus the generation above the minimum level.

$$p_{g,t}^* + r_{g,t}^+ \leq (P_g^{max} - P_g^{min}) \cdot u_{g,t} - (P_g^{max} - SD_g) \cdot w_{g,t+1} - \max(SD_g - SU_g, 0) v_{g,t}, \forall t, g \in \Omega_G \quad (8)$$

$$p_{g,t}^* + r_{g,t}^+ \leq (P_g^{max} - P_g^{min}) \cdot u_{g,t} - \max(SU_g - SD_g, 0) w_{g,t+1} - (P_g^{max} - SU_g) \cdot v_{g,t}, \forall t, g \in \Omega_G \quad (9)$$

$$p_{g,t}^* + r_{g,t}^+ \leq (P_g^{max} - P_g^{min}) \cdot u_{g,t} - (P_g^{max} - SU_g) \cdot v_{g,t} - (P_g^{max} - SD_g) \cdot w_{g,t+1}, \forall t, g \in \Omega_G \quad (10)$$

$$p_{g,t}^* - r_{g,t}^- \geq 0, \forall t, g \in \Omega_G \quad (11)$$

$$p_{g,t} = p_{g,t}^* + u_{g,t} \cdot P_g^{min}, \forall t, g \in \Omega_G \quad (12)$$

d) Ramp rate constraints

Equations (13) and (14) enforce each unit's hourly ramp-up/down capacities. While various ramp rate formulations exist, we employ these specific formulations in this study due to their demonstrated effectiveness in improving the formulation's tightness and reducing CPU time [1].

$$p_{g,t}^* + r_{g,t}^+ - p_{g,t-1}^* \leq RU_g \cdot (u_{g,t} - v_{g,t}) + v_{g,t} \cdot (SU_g - P_g^{min}), t \neq 1, g \in \Omega_G \quad (13)$$

$$p_{g,t-1}^* + r_{g,t}^- - p_{g,t}^* \leq RD_g \cdot (u_{g,t-1} - w_{g,t}) + w_{g,t} \cdot (SD_g - P_g^{min}), t \neq 1, g \in \Omega_G \quad (14)$$

e) System-wide constraints

System-wide constraints include DC power flow Equations (15) and (16), power balance (17), the total upward/downward reserve requirements enforced in (18), and the wind power output limit in (19).

$$f_{l,t} = \sum_{i \in \Omega_B} B_{li} \cdot A_{li}^i \cdot \theta_{i,t} \cdot S_b, \forall t, \forall l \quad (15)$$

$$-F_l^{max} \leq f_{l,t} \leq F_l^{max}, \forall t, \forall l \quad (16)$$

$$\sum_{g \in \Omega_G^i} p_{g,t} + p_{i,t}^w - \sum_{l \in \Omega_l} f_{l,t} \cdot A_{li}^i + LDS_{i,t} = D_t \cdot \alpha_i, \forall t, \forall i \quad (17)$$

$$\sum_{g \in \Omega_G} r_{g,t}^+ \geq RT_t^+, \sum_{g \in \Omega_G} r_{g,t}^- \geq RT_t^-, \forall t \quad (18)$$

$$0 \leq p_{i,t}^w \leq PW_{i,t}, \forall t, i \in \Omega^w. \quad (19)$$

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- [2] G. Morales-España, J. M. Latorre, and A. Ramos, "Tight and compact MILP formulation for the thermal unit commitment problem," *IEEE Trans. Power Syst.*, vol. 28, no. 4, pp. 4897-4908, Nov. 2013.

- [3] J. Ostrowski, M. F. Anjos, and A. Vannelli, "Tight mixed integer linear programming formulations for the unit commitment problem," *IEEE Trans. Power Syst.*, vol. 27, no. 1, pp. 39-46, Feb. 2012.